

SQL Joins

Quick Reference & Study Notes

1. The Four Join Types

Join type	What it returns	Unmatched rows?
INNER JOIN	Only rows with a match in both tables	Dropped entirely
LEFT JOIN	All rows from the left table + matched rows from right	Right side → NULL
RIGHT JOIN	All rows from the right table + matched rows from left	Left side → NULL
FULL OUTER JOIN	All rows from both tables	Opposite side → NULL

2. OUTER is Optional

The **OUTER** keyword is entirely optional in SQL — it is accepted but never required. These pairs are 100% identical:

What you write	What SQL understands
JOIN	INNER JOIN
LEFT JOIN	LEFT OUTER JOIN
RIGHT JOIN	RIGHT OUTER JOIN
FULL JOIN	FULL OUTER JOIN

Rule of thumb: A bare JOIN with no prefix always means INNER JOIN. Most developers omit the OUTER keyword — it's just noise.

3. FULL OUTER JOIN — Database Support

Database	Supports FULL OUTER JOIN?	Notes
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PostgreSQL	✓ Yes	Native syntax, fully supported
SQL Server	✓ Yes	Native syntax, fully supported
Oracle	✓ Yes	Native syntax, fully supported
MySQL / MariaDB	✗ No	Use the UNION workaround (see below)
SQLite	✗ No	Use the UNION workaround

4. MySQL Workaround for FULL OUTER JOIN

Since MySQL does not support FULL OUTER JOIN, combine a LEFT JOIN and a RIGHT JOIN with UNION. The UNION operator automatically removes duplicate rows, producing an identical result:

```
-- MySQL / MariaDB workaround

SELECT *

FROM customers c

LEFT JOIN orders o ON c.id = o.customer_id

UNION

SELECT *

FROM customers c

RIGHT JOIN orders o ON c.id = o.customer_id;
```

Why does UNION work here? The LEFT JOIN captures all customers (including those with no orders). The RIGHT JOIN captures all orders (including orphaned ones with no customer). UNION merges both sets and deduplicates matched rows that appear in both, giving the full picture.

5. Quick Decision Guide

Goal	Use
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Only records that exist in both tables	INNER JOIN
Keep all records from the primary/main table	LEFT JOIN
Keep all records from a secondary table	RIGHT JOIN (or flip + LEFT)
Complete picture of both tables, find gaps	FULL OUTER JOIN

6. Handling NULLs in Outer Joins

Unmatched columns in any outer join return NULL. Use COALESCE to substitute a default value:

```
SELECT  
  
c.name,  
  
COALESCE(o.total, 0) AS order_total,  
  
COALESCE(o.status, 'n/a') AS order_status  
  
FROM customers c  
  
LEFT JOIN orders o ON c.id = o.customer_id;
```

Tip: When filtering a LEFT JOIN result, put conditions on the right-hand table inside the ON clause — not in WHERE. Using WHERE o.status = 'paid' will silently convert your LEFT JOIN into an INNER JOIN because NULL rows fail the WHERE filter.